



One biological pattern, two mathematical lenses

Nutrient-limited growth can destabilise a smooth colony front, while continuum mechanics connects cell-fluid motion to the observed radial expansion. Together, the models explain different parts of the same phenomenon rather than competing descriptions.

The biological question

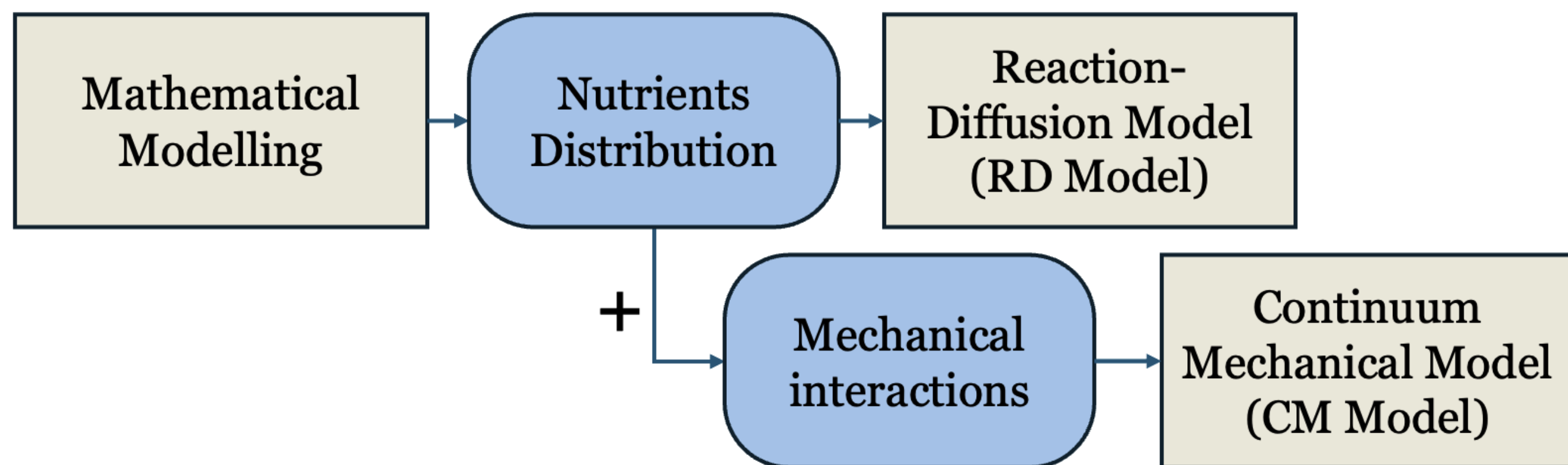
Yeast cells can adhere to a surface, proliferate and produce extracellular matrix (ECM), creating expanding colonies with striking petal-like edges. The central question is:

Which feedbacks select the expansion speed and the characteristic floral pattern?



Budding yeast cells (left) and a floral yeast biofilm (right). Images: Murtey & Ramasamy [5]; Ee Lin Tek, reproduced in Tam [2].

From biology to mathematics



Two complementary routes used in this project. Diagram produced for this project.

	Reaction–diffusion	Continuum mechanics
Focus	nutrient–cell feedback	cell–ECM–fluid motion
Predicts	front speed; unstable modes	radius; expansion speed
Best for	onset and wavelength	bulk colony growth

Scope and contribution

This is a **critical synthesis of published models** [1,2], not a claim that the group generated new experimental data. Reproduced plots are labelled explicitly; the contribution here is to connect assumptions, equations, numerical methods and biological interpretation in one coherent framework.

References and image credits

- [1] A. Tam et al., “Nutrient-limited growth with non-linear cell diffusion as a mechanism for floral pattern formation in yeast biofilms,” *J. Theor. Biol.* 448 (2018), 122–141. doi:10.1016/j.jtbi.2018.04.004.
- [2] A. Tam, *Mathematical Modelling of Pattern Formation in Yeast Biofilms*, doctoral thesis, University of Adelaide (2019).
- [3] T. B. Reynolds and G. R. Fink, “Bakers’ yeast, a model for fungal biofilm formation,” *Science* 291 (2001), 878–881.
- [4] J. Talapko et al., “Candida albicans—the virulence factors and clinical manifestations of infection,” *J. Fungi* 7 (2021), 79.
- [5] M. D. Murtey and P. Ramasamy, “Sample preparations for scanning electron microscopy—life sciences,” IntechOpen (2016).

Model figures are reproduced/adapted from [1,2] under the licences stated in the source material.

MODEL 1 Nutrient-limited reaction–diffusion

Density-dependent cell spreading and nutrient consumption are represented by

$$\begin{aligned}\frac{\partial n}{\partial t} &= D_n \nabla \cdot (n \nabla n) + png, \\ \frac{\partial g}{\partial t} &= D_g \nabla^2 g - cpng.\end{aligned}$$

n : cell density g : nutrient concentration D_n, D_g : diffusivities
 p : growth coupling c : nutrient-consumption factor

The nonlinear flux $\nabla \cdot (n \nabla n)$ makes spreading density dependent: cells diffuse only where the population is present. The same png term adds cells while $cpng$ removes nutrient [1].

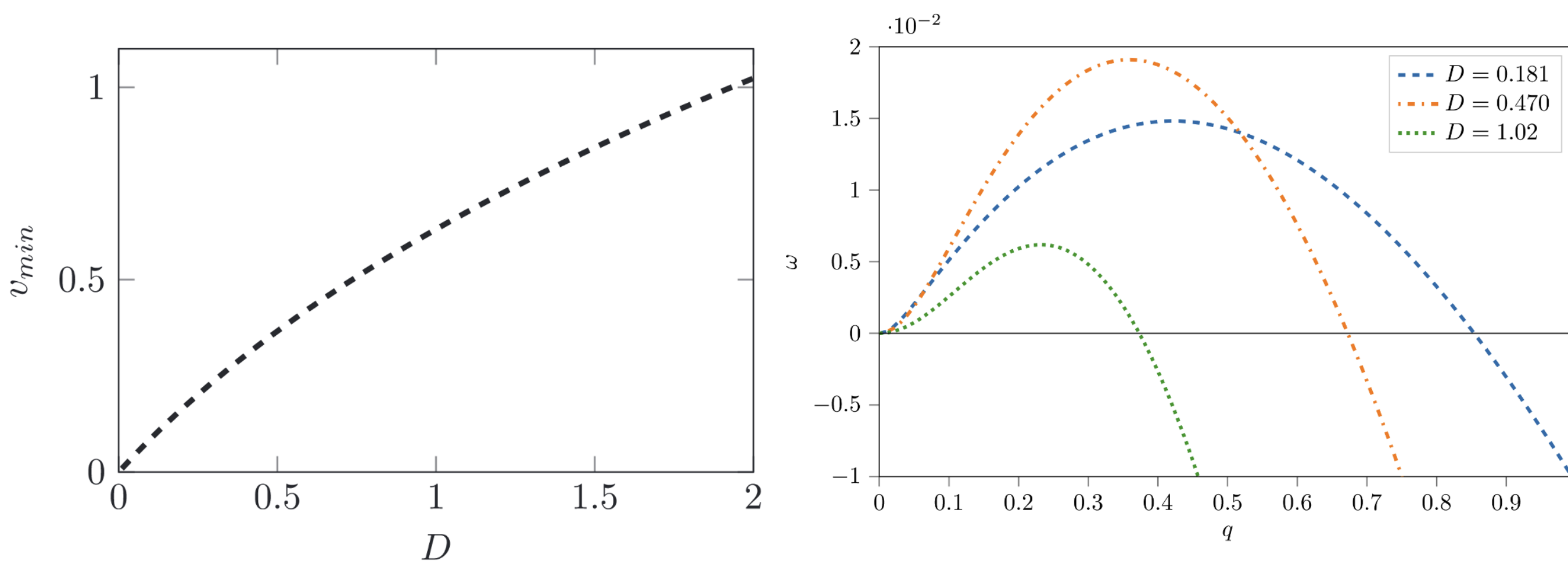
How a smooth front becomes floral

- 1 A travelling-wave solution describes uniform radial expansion.
- 2 Perturb the front by an angular mode with wavenumber q .
- 3 Linear stability gives a dispersion relation $\omega(q)$.

$\omega(q) > 0$ means that the perturbation grows. The fastest-growing mode q^* selects a characteristic petal spacing

$$\lambda^* = \frac{2\pi}{q^*}$$

What the analysis predicts



Left: minimum front speed $v_{\min}$ versus the reduced diffusivity parameter D . Right: dispersion curves $\omega(q)$ for three values of D . Reproduced from Tam [2], based on [1].

- Increasing D raises the predicted minimum expansion speed.
- A band of modes with $\omega > 0$ is unstable; changing D shifts both the growth rate and the selected wavenumber.
- Petal spacing is a spatial wavelength, $\lambda = 2\pi/q$; it should not be confused with the temporal growth rate ω .

Numerics and verification

Travelling-wave and stability calculations are complemented by second-order finite differences, Crank–Nicolson time stepping and iterative linear solves [1]. A credible result requires **grid and time-step refinement**: conclusions should remain unchanged as Δx and Δt decrease.

Reproducibility test. Report $Q \in \{v_{\min}, q^*\}$ on h , $h/2$ and $h/4$. For second-order refinement, $R_Q = |Q_h - Q_{h/2}|/|Q_{h/2} - Q_{h/4}| \approx 4$. Repeat for Δt and keep solver residuals below discretisation error.

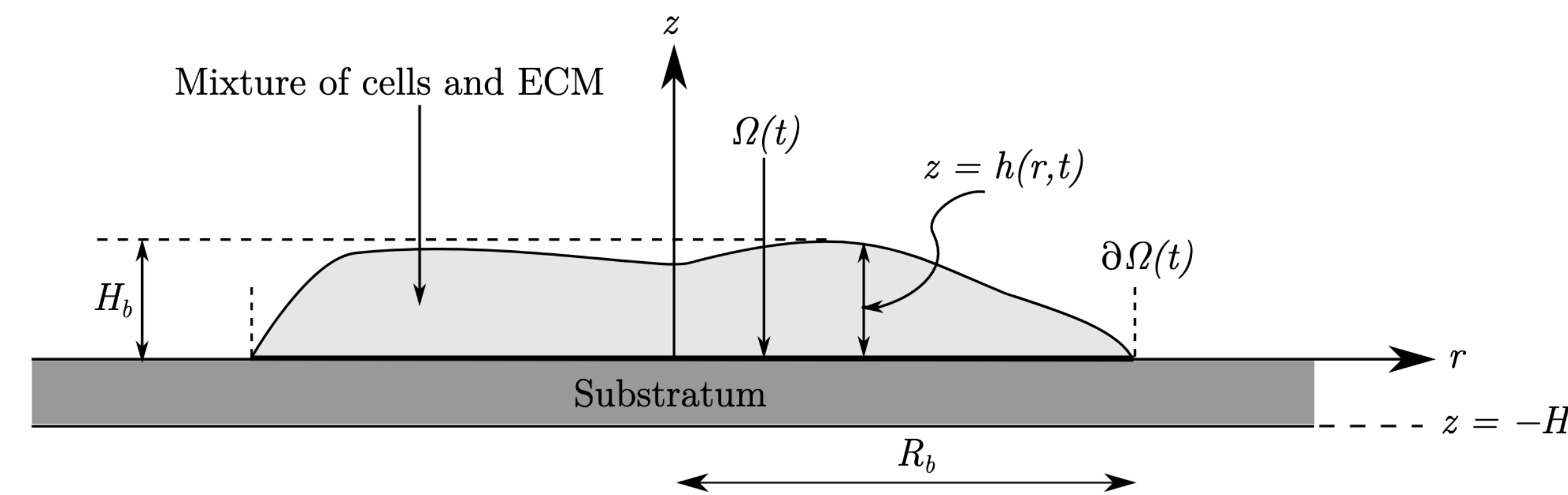
MODEL 2 Continuum mixture mechanics

Treat cells and ECM as interacting phases. For phase α , mass and momentum balances take the generic form

$$\begin{aligned}\rho_\alpha \left[\frac{\partial \phi_\alpha}{\partial t} + \nabla \cdot (\phi_\alpha \mathbf{u}_\alpha) \right] &= J_\alpha, \\ \rho_\alpha \frac{D}{Dt} (\phi_\alpha \mathbf{u}_\alpha) &= \nabla \cdot (\phi_\alpha \boldsymbol{\sigma}_\alpha) + \mathbf{F}_\alpha.\end{aligned}$$

Here ϕ_α is volume fraction, $\mathbf{u}_\alpha$ velocity, $\boldsymbol{\sigma}_\alpha$ stress, J_α a mass source and $\mathbf{F}_\alpha$ interphase/body force. At biofilm scales inertia is negligible, so the left-hand side of the momentum equation is reduced accordingly [2].

Thin-film reduction and adhesion regimes



Axisymmetric biofilm geometry and moving contact line. Adapted from Tam [2].

For $\varepsilon = H_b/R_b \ll 1$, a thin-film expansion resolves vertical structure asymptotically and produces a depth-averaged free-boundary model; it does **not** simply “ignore the z -axis.” Axisymmetry then leaves one radial coordinate.

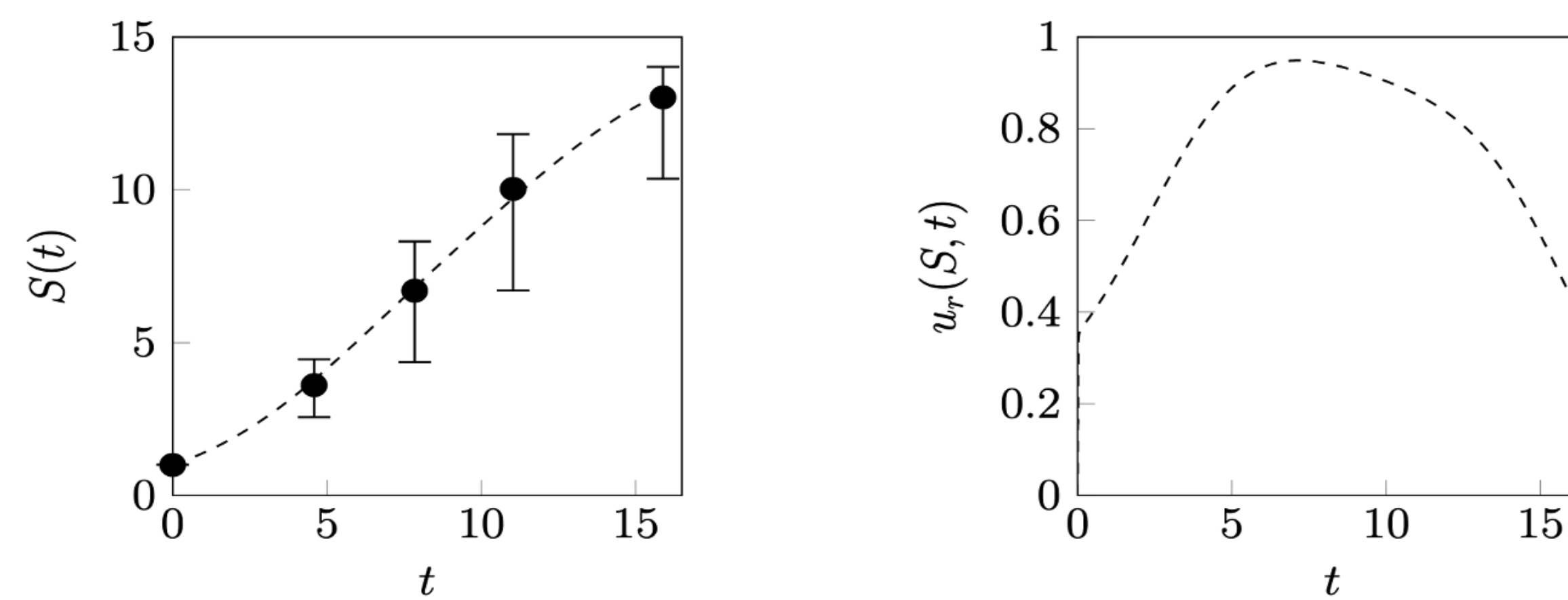
Weak substrate adhesion

Extensional-flow regime
sliding-driven expansion

Strong substrate adhesion

Lubrication regime
no-slip, pressure and capillarity

Model–experiment comparison



Left: predicted contact-line radius $S(t)$ with measurements from 13 experiments. Right: predicted radial velocity profile. Reproduced from Tam [2].

The predicted mean radius is consistent with the reported experimental uncertainty. The moving domain is mapped by $\xi = r/S(t)$ before time stepping and refinement checks [2]. Agreement supports the fitted mechanism; it does **not** prove uniqueness or universal accuracy.

Take-home messages

1. Nutrient feedback can select unstable floral length scales.
2. Mechanics maps growth and adhesion to colony radius and speed.
3. Joint measurements of wavelength and radial dynamics would test both models.